

The CKM Quark Mixing Matrix from Foam Cell Geometry

Cabibbo Angle, Wolfenstein Parameters, and the Froggatt-Nielsen Texture

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Framework: Unified Foam Field Theory (UFFT)

Abstract

Two of the four Wolfenstein parameters of the CKM quark mixing matrix are derived from the geometry of the truncated octahedron. The Cabibbo angle is $\theta_C = \pi/F$ where $F = 14$ is the face count: $\sin(\pi/14) = 0.22252$, matching the observed $\lambda = 0.22500 \pm 0.00067$ to 1.1% (within 0.4σ). The second parameter $A = r_1/C_A = (9-\sqrt{17})/6 = 0.8128$, where r_1 is the smaller root of the master equation $\lambda^2 - 9\lambda + 16 = 0$ and $C_A = 3$ is the colour number; this matches the observed $A = 0.826 \pm 0.015$ to 1.6% (within 0.9σ). The derivation reveals a new connection: the down-to-strange quark mass ratio $m_d/m_s \approx \sin^2(\pi/14)$ to 1%, providing a geometric origin for the Froggatt-Nielsen texture — the empirical observation that quark mass hierarchies scale as powers of the Cabibbo angle. From these two parameters, $|V_{cb}| = A\lambda^2 = 0.0402$ (observed: 0.0412, 2.4%). The CP-violating parameters $\bar{\rho}$ and $\bar{\eta}$ remain open — they require the complex phase of the inter-generation torsion coupling. Zero free parameters.

Keywords: CKM matrix, Cabibbo angle, quark mixing, Wolfenstein parameters, Froggatt-Nielsen, quark mass hierarchy, truncated octahedron, face adjacency, foam field theory, UFFT

1. The CKM Matrix

The Cabibbo-Kobayashi-Maskawa matrix V_{CKM} rotates between quark mass eigenstates and weak interaction eigenstates [1,2]. In the Wolfenstein parametrisation [3]:

$$V_{CKM} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\bar{\rho}-i\bar{\eta}) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1-\bar{\rho}-i\bar{\eta}) & -A\lambda^2 & 1 \end{pmatrix}$$

Four parameters: λ (Cabibbo angle), A (2→3 generation suppression), $\bar{\rho}$ and $\bar{\eta}$ (CP violation). Current experimental values [4]: $\lambda = 0.22500 \pm 0.00067$, $A = 0.826 \pm 0.015$, $\bar{\rho} = 0.159 \pm 0.010$, $\bar{\eta} = 0.348 \pm 0.010$.

No part of the Standard Model determines these four numbers. They are measured, not calculated. Their values have no known theoretical origin.

2. The Cabibbo Angle from the Face Count

2.1 Physical mechanism

In the UFFT framework, quark mass eigenstates are determined by the T₂g torsion sector of the truncated octahedral foam (eigenvalue λ = 7). Weak interaction eigenstates are determined by the E_g chiral discharge sector (eigenvalue λ = 4). The CKM matrix is the rotation between these two bases [5].

The truncated octahedron has F = 14 faces: 6 squares and 8 hexagons. Each face is a boundary between adjacent cells. The torsion and discharge modes propagate through different subsets of faces. The minimal angular mismatch between the two mode propagation directions, quantised by the discrete face geometry, is:

$\theta_C = \pi/F = \pi/14$

2.2 Numerical result

Quantity	UFFT	Observed	Accuracy
$\sin \theta_C = \lambda$	$\sin(\pi/14) = 0.22252$	0.22500 ± 0.00067	1.1% (0.4σ)
θ_C	12.857°	13.00°	1.1%

3. The Parameter A from the Master Equation

3.1 Physical mechanism

The parameter A governs the suppression of 2nd-to-3rd generation transitions ($|V_{cb}| = A\lambda^2$). In the foam, the three quark generations correspond to the three orientations of the T₁u torsion-displacement coupling mode. The coupling between the 2nd and 3rd generations involves the smaller T₁u eigenvalue r₁ = (9-√17)/2, normalised by the colour number C_A = 3 (because colour-singlet hadrons require averaging over C_A colour states).

$A = r_1/C_A = (9-\sqrt{17})/6$

3.2 Numerical result

Quantity	UFFT	Observed	Accuracy
A	$(9-\sqrt{17})/6 = 0.8128$	0.826 ± 0.015	1.6% (0.9σ)

4. Derived CKM Elements

From the two foam parameters:

Element	Formula	UFFT	Observed	Accuracy
V_us	$\sin(\pi/14)$	0.2225	0.2250	1.1%
V_cd	$\sin(\pi/14)$	0.2225	0.2250	1.1%
V_cb	$(9-\sqrt{17})/6 \times \sin^2(\pi/14)$	0.0402	0.0412	2.4%
V_ts	$(9-\sqrt{17})/6 \times \sin^2(\pi/14)$	0.0402	0.0404	0.5%

The elements |V_ub| and |V_td| involve the CP-violating parameters $\bar{\rho}$ and $\bar{\eta}$, which are not yet derived.

5. The Froggatt-Nielsen Connection

5.1 The empirical pattern

Froggatt and Nielsen [6] observed that quark mass ratios within each charge sector scale approximately as powers of the Cabibbo angle:

$m_d/m_s \approx \lambda^2, \quad m_u/m_c \approx \lambda^4$

This pattern has been used extensively in flavour model-building but has never been derived from first principles.

5.2 The foam derivation

With $\lambda = \sin(\pi/14)$:

$m_d/m_s \approx \sin^2(\pi/14) = 0.0495$

Observed: $m_d/m_s = 4.67/93.4 = 0.0500$

Accuracy: 1.0%

This is not a coincidence. The face count $F = 14$ determines both the Cabibbo angle (the angular mismatch between mass and weak eigenstates) AND the mass hierarchy (the eigenvalue spacing between generations). Both arise from the same geometric structure: the 14-face boundary of the Kelvin cell.

The physical mechanism: the mass ratio between adjacent generations is suppressed by the square of the mixing angle because mass generation requires two torsion interactions (one at each end of the quark propagator), each carrying a factor of $\sin(\pi/F)$.

6. What Remains Open

6.1 The CP-violating parameters

The parameters $\bar{\rho} = 0.159$ and $\bar{\eta} = 0.348$ determine the CP-violating phase $\delta_{\text{CKM}} \approx 65^\circ$. In the foam, this phase arises from the complex structure of the inter-generation torsion coupling. The truncated octahedron's dihedral angle supplements (54.7° for square-hexagon, 70.5° for hexagon-hexagon, average 62.6°) are in the vicinity of the observed 65° but no clean derivation has been found.

The Jarlskog invariant $J = A^2 \lambda^6 \bar{\eta} \approx 3.08 \times 10^{-5}$ requires $\bar{\eta}$ and therefore remains open.

6.2 The higher-generation mass ratios

While $m_d/m_s \approx \lambda^2$ matches to 1%, the analogous relations $m_s/m_b \approx \lambda^2$ (predicts 0.050, observed 0.022, off by 2.2 $\times$) and $m_u/m_c \approx \lambda^4$ (predicts 0.0025, observed 0.0017, off by 44%) are less accurate. The full quark mass spectrum requires the complete knot classification programme.

7. Summary

Parameter	Foam formula	Value	Observed	Status
λ	$\sin(\pi/F)$	0.22252	0.22500	DERIVED (1.1%)
A	r_1/C_A	0.8128	0.826	DERIVED (1.6%)
$\bar{\rho}$	—	—	0.159	OPEN
$\bar{\eta}$	—	—	0.348	OPEN
m_d/m_s	$\sin^2(\pi/F)$	0.0495	0.0500	DERIVED (1.0%)

Three results from one cell integer ($F = 14$) and one spectral root (r_1). Zero free parameters.

References

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AI Disclosure

This paper was developed in collaboration with Claude (Anthropic). Ideas, direction, and framework: Luke Martin. AI role: numerical exploration, derivation formulation, document composition.

UFFT Core Framework: github.com/WebEnvy/UnifiedFoamFieldTheory